Math 5601: Introduction to Numerical Analysis
Homework assignment 8

Show all relevant work in detail to justify your conclusions. Partial credit depends upon the work you show.

Problem #1: Consider the formula

$$\int_0^h f(x)\,dx \approx h\left[Af(0) + Bf\left(\frac{h}{3}\right) + Cf(h)\right].$$

(a) Find A,B,and C such that this is exact for all polynomials of degree less than or equal to 2.

(b) Suppose that the trapezoidal rule applied to $\int_0^2 f(x)\,dx$ gives the value $\frac{1}{2}$ while the quadrature rule in part (a) applied to $\int_0^2 f(x)\,dx$ gives the value $\frac{1}{4}$. If $f(0) = 3$, then show that $f\left(\frac{2}{3}\right) = 1$.

Problem #2: Show that

(a) forward phase of Gauss elimination requires $\frac{1}{3}n^3 + O(n^2)$ multiplications and divisions.

(b) back solving Gauss elimination requires $\frac{n^2+n}{2}$ multiplications and divisions.

$\sqrt{4} =$

$\rightarrow x^2 - 4 = 0$

$\rightarrow x = \pm 4$

$\rightarrow x = \pm 2$

4
2

-4
-2

1 + (-7)

Math 5601 - Homework 8 - Tyler Glidden

Problem 1

a.) $\int_0^h f(x)\,dx \approx h\left[A f(0) + B f(h/3) + C f(h)\right]$

Find A, B, C St this is exact for polynomials of degree ≤ 2

$x_0 = 0$

$x_1 = h/3$

$x_2 = h$

$a_0 \Rightarrow h(A+B+C) = \int_0^h 1\,dx = x\Big|_0^h = h$ //

$a_1 \Rightarrow h(Ax_0 + Bx_1 + Cx_2) = \int_0^h x\,dx = \frac{x^2}{2}\Big|_0^h = \frac{h^2}{2}$ //

$a_2 \Rightarrow h(Ax_0^2 + Bx_1^2 + Cx_2^2) = \int_0^h x\,dx = \frac{h^3}{3}$ //

$\int_0^h f(x)dx = h\left[\frac{3}{4} f(h/3) + \frac{1}{4} f(h)\right]$

$$\begin{cases} h(A+B+C) = 1 \cdot h \\ h(A(0) + B\,h/3 + Ch) = h^2/2 \\ h\left(A(0) + B\,h^2/9 + Ch^2\right) = h^3/3 \end{cases}$$

$$\begin{cases} A+B+C = 1 \\ \frac{B}{3} + C = 1/2 \\ B/9 + C = 1/3 \end{cases}$$

$A = 0$

$B = 3/4$

$C = 1/4$ //

b.) Suppose Trapezoidal gives $\int_0^2 f(x)dx$, $1/2$. and Part a.) gives $1/4$

If $f(0) = 3$, show $f(2/3) = 1$

Trapezoidal:

$\int_a^b f(x)dx = \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b)$

$\int_0^h f(x)dx = \frac{h-0}{2} f(0) + \frac{h-0}{2} f(h)$

Given $\int_0^2 f(x)dx = \frac{2}{2} f(0) + \frac{2}{2} f(2) = \frac{1}{2}$

Part a)

$\int_0^h f(x)dx = h\left[\frac{3}{4} f(h/3) + \frac{1}{4} f(h)\right]$

$\int_0^2 f(x)dx = 2\left[\frac{3}{4} f(2/3) + \frac{1}{4} f(2)\right] = 1/4$

use:

$$\begin{cases} f(0) + f(2) = 1/2 \Rightarrow f(2) = 1/2 - f(0) \Rightarrow f(2) = 1/2 - 3 = -5/2 \text{ //} \\ 6/4\, f(2/3) + 2/4\, f(2) = 1/4 \Rightarrow \frac{6}{4} f(2/3) + \frac{2}{4}(-5/2) = 1/4 \Rightarrow 6 f(2/3) - 5 = 1 \end{cases}$$

$6 f(2/3) = 6$

$\underline{f(2/3) = 1}$ ■

Problem 2

a.) Forware Phase Gauss

$\frac{1}{3}n^3 + O(n^2)$

For $A = n \times n$, $\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & & \vdots \\ \vdots & & \\ a_{n1} & & a_{nn} \end{bmatrix}$

To Find m_{k1}

$m_{21} = \frac{a_{21}}{a_{11}}$ $m_{31} = \frac{a_{31}}{a_{11}}$ ... $m_{n1} = \frac{a_{n-1}}{a_{11}}$

$(n-k)$ Divisions

To make zeros below a_{11}, $k=1$,

Row 2

$a_{21} = 0$, no calculation

$a_{22} = -m_{21} a_{22} + a_{22}$

$\vdots$

$a_{2n} = -m_{2n} a_{2n} + a_{2n}$

$\rightarrow (n-k)$

Row 3

$\vdots$

$(n-k)$ $\rightarrow (n-k)$ Times, so $(n-k)(n-k)$

Row 4

$(n-k)$

$\vdots$

Row n

$(n-k)$

For step 1

$(n-k) + (n-k)^2$ steps,

now we need to sum

$\sum_{k=1}^{n-1} (n-k) + (n-k)^2 \rightarrow$ total Rows.

$= \sum_{k=1}^{n-1} n^2 - 2kn + k^2 + n - k$

$= \sum_{k=1}^{n-1} n^2 - 2n\sum_{k=1}^{n-1} k + \sum_{k=1}^{n-1} k^2 + \sum_{k=1}^{n-1} n - \sum_{k=1}^{n-1} k$

$(n-1)\ (2n^2 \cdot 2)$

$(n-1)n^2 - \frac{2n(n-1)n}{2} + \frac{(n-1)n(2n-1)}{6} + n(n-1) - \frac{(n-1)n}{2}$

$n^3 - n^2 - \frac{2n^3}{2} + \frac{2n^2}{2} + \frac{2n^3}{6} - 2n - \frac{2n^2}{2} + \frac{n}{2} + n^2 - n - \frac{n^2}{2} + \frac{n}{2}$

$\frac{2n^3}{6} + O(n^2)$

$= \frac{1}{3}n^3 + O(n^2)$

$\sum_{i=1}^{n-1} i = \frac{(n-1)\cdot n}{2}$ $\sum_{i=1}^{n-1} i^2 = \frac{(i-1)\,i(2i-1)}{6}$

b.) Back Solving Gauss

A is now upper diagonal

mult. and divisions: $\frac{n^2+n}{2}$

$$[A \vdots \vec{b}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & \vdots & b_1 \\ & a_{22} & & a_{2n} & \vdots & b_2 \\ & & \ddots & \vdots & \vdots & \vdots \\ & 0 & & a_{nn} & \vdots & b_n \end{bmatrix}$$

For X_n (Step 1)

$X_n = b_n / a_{nn}$ → 1 division + 0 mult. = 1

For X_{n-1} (Step 2)

$a_{n-1,n-1} X_{n-1} + \underbrace{a_{n-1,n} \cdot X_n}_{w} = b_{n-1}$ → 1 division + 1 multiplication = 2

1 mult

$$X_{n-1} = \frac{b_{n-1} - w}{a_{n-1}}$$

b_{n-1}

⋮

For X_1 (Step n)

$a_{11}X_1 + a_{12}X_2 + \ldots + a_{1n}X_n = b_1$

w — $n-1$ multiplications

1 div + $n-1$ mult. = n

sum n rows

$$\frac{a_{11}X_1}{a_{11}} = \frac{b_1 - w}{a_{11}}$$

$$X_1 = \frac{b_1 - w}{a_{11}}$$ ← 1 division

$$\sum_{i=1}^{n} n = \frac{(n+1)n}{2} = \frac{n^2+n}{2}$$

"Inspiration"

on next Page →

$$A = \begin{bmatrix} 1 & 3 & 1 & 4 \\ 3 & 9 & 5 & 15 \\ 0 & 2 & 1 & 1 \\ 0 & 4 & 2 & 3 \end{bmatrix}$$

$$m_{i1} = \frac{3}{1}, \frac{0}{1}, \frac{0}{1} = \frac{a_{i1}}{a_{11}} \quad \text{for } i = 2,3,4$$

$$m_{i2} = \frac{"2"}{9}, \frac{"4"}{9} = \frac{a_{i2}}{a_{22}} \quad \text{for } i = 3,4$$

$$m_{i3} = \frac{2}{1} = \frac{a_{i3}}{a_{33}} \quad \text{for } i = 4$$

To kill column 1

Row 2

① $a_{21} = -m_{21} \cdot a_{11} + a_{21} = -3 \cdot 1 + 3 = 0$ ← no need to calc. set = 0

② $a_{22} = -m_{21} \cdot a_{12} + a_{22} = -3 \cdot 3 + 3 = -6$

③ $a_{23} = -m_{21} \cdot a_{13} + a_{23} = -3 \cdot 1 + 5 = 2$

④ $a_{24} = -m_{21} \cdot a_{14} + a_{24} = -3 \cdot 4 + 15 = 3$

Row 3

① $a_{31} = -m_{31} \cdot a_{11} + a_{31} = 0$

②

③

④

Row 4

① $= 0$

②

③

④

To kill column 2

Row 3

① $= 0$

②

③

Row 4

① $= 0$

②

③

To kill Column 3

Row 4

① $= 0$

②

$$\begin{bmatrix} 1 & 3 & 1 & 4 \\ 0 & -6 & 2 & 3 \\ 0 & x & x & x \\ 0 & x & x & x \end{bmatrix}$$

to kill row k

$(n-k)$ divisions

Update Row k

$(n-k)(n-k)$

$(n-k)^2 + (n-k)$

$$\sum_{1}^{n-1} (n-k)^2 + (n-k)$$

For

How to do sums??

101.

$$\frac{(n-k)(n-k+1)(2(n-k)+1)}{6} - (n-k)$$

$1 + \ldots + 100$

$1^2 + \ldots + 99^2 (+ 100^2)$

$$\sum_{i=1} i^2 =$$

$$\sum_{i=1}^{99} i^2 = \sum_{i=1}^{100} i^2 - 100^2 =$$

n-1	4-1
n-2	4-2
n-3	4-3
n-4	4-4

3

3

3

2

1

$3 \cdot 3 + 2 \cdot 2 + 1 \cdot 1$

$$[A:\vec{b}] = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ & a_{22} & & a_{2n} & b_2 \\ & & \ddots & \vdots & \vdots \\ & 0 & & a_{nn} & b_n \end{pmatrix}$$

$$\begin{pmatrix} 1 & 6 & -3 & 7 & -6 \\ & 1 & 2 & 4 & 3 \\ & & 3 & 1 & 2 \\ & & & 4 & 8 \end{pmatrix}$$

For row 4

$X_4 = 8/4$ ①

For Raw 3

$2X_3 + 1X_4 = 2$ ②

$2X_3 + 1 \cdot 2 = 2$

$2X_3 = 0$

$X_3 = 0/2$ ①

For Low 2

$2X_2 + 2X_3 + 4X_4 = 3$ ③

$2X_2 + 2 \cdot 0 + 4 \cdot 2 = 3$

① ②

$2X_2 = 3 - 6$

$X_2 = -9$